Stochastic Simulations

Stochastic - not predictable in advance, degree of randomness
Stochastic process - sequence of random events

Poisson Distribution - random # events happening in time period

Random var is distributed according to Poisson distribution with rate λ

$$X \sim \text{Poisson}(\lambda)$$

must be >0 must be whole number

Binomal Distribution → $X \sim \text{Binomal}(k, p)$

x = # successes
k = # trials
p = probability in trial resulting in success

Normal Distribution - $X \sim \text{Normal}(m, s)$
x = value
m = mean
s = std

Set.seed(x) - keeps random was generated